

Replicating Chetty et al. (2014): Classical vs Permutation Inference for Intergenerational Mobility

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Introduction

Background: The Chetty et al. Study

In their landmark 2014 paper “Where is the Land of Opportunity? The Geography of Intergenerational Mobility in the United States,” Chetty, Hendren, Kline, and Saez analyzed tax records from over 40 million children to measure intergenerational economic mobility across U.S. Commuting Zones (CZs).

A key finding was identifying which community-level characteristics correlate with **relative mobility**—the relationship between a child’s income rank and their parents’ income rank. Online Appendix Table VIII of their paper reports correlations between relative mobility and 35 covariates spanning categories like segregation, income inequality, education, family structure, and social capital.

Purpose of This Analysis

This analysis has two objectives:

1. **Replicate** the classical OLS regression results from Chetty et al.’s Online Appendix Table VIII
2. **Compare** these classical results to a permutation-based inference method that does not rely on asymptotic normality assumptions

Why Permutation Tests?

Classical OLS regression relies on asymptotic theory—as sample sizes grow large, the t-statistic approximately follows a t-distribution. However, this approximation may be unreliable when:

- Sample sizes are moderate (here, $n \approx 700$ CZs)
- Error distributions are non-normal or heteroskedastic
- The relationship between variables violates model assumptions

Permutation tests offer an alternative that:

- Makes no distributional assumptions about the errors
- Provides exact p-values under the null hypothesis of no relationship
- Is robust to heteroskedasticity when properly studentized

We implement the permutation regression test from **DiCiccio and Romano (2017)**, “Robust Permutation Tests for Correlation and Regression Coefficients,” which provides a heteroskedasticity-robust test statistic.

The Data: Tables 5 and 8

Our analysis uses two of Chetty et al.’s Online Data Tables:

Online Data Table 5 provides the **dependent variable**: relative mobility estimates for 741 Commuting Zones (CZs). These estimates were computed from IRS tax records covering over 40 million children born between 1980-1991, linked to their parents’ tax returns. The key variable is the **rank-rank slope**—the regression coefficient when regressing children’s income percentile rank on parents’ income percentile rank within each CZ. A higher slope indicates less mobility (children’s outcomes more strongly determined by parents’ income). The national average is 0.34, meaning a 10 percentile increase in parent rank is associated with a 3.4 percentile increase in child rank. We use the 1980-82 birth cohort estimates and restrict to CZs with at least 250 children to ensure reliable estimates.

Online Data Table 8 provides the **independent variables**: 35 community-level characteristics for each CZ, including measures of segregation, income inequality, school quality, social capital, family structure, and local labor market conditions. These covariates come from public sources (Census, IRS public statistics, education databases) and allow us to ask: “Which community characteristics predict higher or lower mobility?”

Why Replicate Table VIII?

We chose to replicate Online Appendix Table VIII because it represents the core empirical contribution of Chetty et al. (2014): identifying which community-level characteristics are associated with intergenerational mobility. While the mobility estimates in Table 5 were computed from restricted IRS tax records and cannot be independently reproduced from raw data, Table VIII’s correlational analysis uses these *published* mobility estimates as the dependent variable and publicly available community characteristics as predictors. This makes it both replicable and substantively important—it is this analysis that identifies single-parent household rates as “the single strongest correlate of upward income mobility among all the variables we explored” (Chetty et al., 2014, p. 40), and that links segregation, inequality, and social capital to children’s economic outcomes. Policy discussions citing this paper typically draw on Table VIII’s findings, making it essential to verify that these correlations are robust to alternative statistical methods.

A Note on P-Values and Hypothesis Testing

A p-value greater than 0.05 **never** leads to rejecting the null hypothesis when using $\alpha = 0.05$ as the significance threshold. By definition, we reject H_0 only when $p < \alpha$. When $p \geq 0.05$, we “fail to reject” the null—this is not the same as accepting the null or proving no effect exists; it simply means the observed data are not sufficiently improbable under H_0 to warrant rejection at our chosen confidence level.

Some important clarifications:

- **Different α levels:** If a researcher uses $\alpha = 0.10$, then $p = 0.07$ would lead to rejection. The choice of α is made before seeing the data.
- **One-tailed vs. two-tailed tests:** A two-tailed $p = 0.08$ corresponds to a one-tailed $p = 0.04$, which would reject under a one-tailed test. However, our analysis uses two-tailed tests throughout.
- **Multiple testing:** When testing many hypotheses (as we do with 35 covariates), some researchers apply corrections like Bonferroni ($\alpha_{adjusted} = 0.05/35 \approx 0.0014$), which would make rejection even more stringent.

In this analysis, we use the conventional $\alpha = 0.05$ two-tailed threshold without multiple testing correction, consistent with Chetty et al.’s original analysis.

Research Question

Do the significance conclusions from Chetty et al.’s classical OLS analysis hold up under permutation-based inference?

If both methods agree, this provides evidence that the original findings are robust to the choice of inference method.

Methods

Data Sources

We use two publicly available datasets from the Equality of Opportunity Project:

Dataset	Contents
Online Data Table 5	Mobility estimates (relative and absolute) for each CZ
Online Data Table 8	CZ-level characteristics (35 covariates)

Following Chetty et al., we restrict the sample to CZs with at least 250 children ($n \geq 250$) to ensure reliable mobility estimates.

Statistical Methods

Method 1: Classical OLS Regression (Chetty et al.’s Approach)

For each covariate X_j , we estimate:

$$\text{Relative Mobility}_i = \alpha + \beta_j X_{ji} + \epsilon_i$$

where both the outcome and covariate are standardized (mean 0, SD 1). The standardized coefficient β_j equals the Pearson correlation coefficient r .

Statistical significance is assessed using the classical t-test, which assumes $\epsilon_i \sim N(0, \sigma^2)$.

Method 2: Permutation Correlation Test (perk package)

The **perk** package implements permutation tests for Pearson correlation. For each covariate, we:

1. Compute the observed Pearson correlation r between the covariate and relative mobility
2. Permute the outcome variable Y many times ($B = 10,000$) to generate a null distribution
3. Compute the p-value as the proportion of permuted correlations with $|r^*| \geq |r_{obs}|$

This tests $H_0 : \rho = 0$ (no correlation) without assuming bivariate normality.

Method 3: Permutation Regression Test (DiCiccio & Romano, 2017)

The permutation regression test assesses significance by:

1. Computing a heteroskedasticity-robust Wald statistic S_n from the observed data
2. Permuting the outcome variable Y many times ($B = 10,000$) to generate a null distribution
3. Computing the p-value as the proportion of permuted statistics $\geq$ the observed statistic

The test statistic is:

$$S_n = n\hat{\beta}^T (\hat{\Sigma}_{XX}^{-1} \hat{\Omega} \hat{\Sigma}_{XX}^{-1}) \hat{\beta}$$

where $\hat{\Omega}$ is the heteroskedasticity-robust covariance estimator. This follows Theorem 3.2 of DiCiccio and Romano (2017).

Key advantages:

- No assumption of normally distributed errors
- Robust to heteroskedasticity
- Exact Type I error control under the null hypothesis

Why Both Permutation Tests?

We include both **perk** (permutation correlation) and the DiCiccio & Romano regression test to compare:

1. **perk**: Uses the raw correlation coefficient as the test statistic
2. **DiCiccio & Romano**: Uses a heteroskedasticity-robust Wald statistic

For standardized data, both test the same null hypothesis ($\beta = 0$), but they may yield different p-values due to different test statistics. Comparing them helps assess sensitivity to the choice of permutation method.

Console Output Format

Results are printed in the following format:

```
[OK] Racial.Segregation  b=0.406 p_class=0.000 p_perk=0.000 p_reg=0.000*
```

Symbol	Description
[OK]	All three methods agree on significance
[X]	At least one method disagrees
b	Standardized regression coefficient (= Pearson r)
p_class	p-value from classical OLS t-test
p_perk	p-value from perk permutation correlation test
p_reg	p-value from DiCiccio & Romano permutation regression test
*	Statistically significant at $\alpha = 0.05$

Part 1: Data Loading and Preparation

Load Required Libraries

```
library(tidyverse)
library(broom)
library(perk)
library(sandwich) # For vcovHC (heteroskedasticity-consistent covariance)
library(lmtest)   # For waldtest
library(tseries)  # For jarque.bera.test
```

Load the Chetty et al. Data

The data files are structured with metadata rows at the top, which we skip during import.

```
# Load Online Data Table 5 (Mobility Estimates)
# Row 50 contains column headers, so skip first 49 rows
table5 <- read.csv("../data/Online_Data_Table_5.csv",
  skip = 49,
  header = TRUE,
  stringsAsFactors = FALSE
)

# Remove rows with column numbers "(1)", "(2)", etc. and empty rows
table5 <- table5[!grepl("^\\(\\d+\\)$", table5[, 1]) & table5[, 1] != "", ]
rownames(table5) <- NULL

# Load Online Data Table 8 (CZ Characteristics)
# Row 7 contains column headers, so skip first 6 rows
table8 <- read.csv("../data/Online_Data_Table_8.csv",
  skip = 6,
  header = TRUE,
  stringsAsFactors = FALSE
)

# Remove invalid rows (negative CZ codes are metadata)
table8 <- table8[as.numeric(table8[, 1]) > 0 & !is.na(table8[, 1]), ]
rownames(table8) <- NULL

if (nrow(table8) == 0) {
  stop("ERROR: Online_Data_Table_8.csv loaded 0 rows! Check file location.")
}

cat(sprintf("Loaded Table 5: %d CZs with mobility estimates\n", nrow(table5)))
```

Loaded Table 5: 741 CZs with mobility estimates

```
cat(sprintf("Loaded Table 8: %d CZs with characteristics\n", nrow(table8)))
```

Loaded Table 8: 741 CZs with characteristics

Clean and Prepare Data

```
# Clean column names (R converts spaces and special chars to dots)
names(table5) <- gsub("^X\\.", "", names(table5))
names(table8) <- gsub("^X\\.", "", names(table8))

# Rename key columns for clarity
```

```

colnames(table5)[1] <- "CZ"
colnames(table5)[5] <- "AM"           # Absolute Mobility
colnames(table5)[6] <- "RM"           # Relative Mobility
colnames(table5)[4] <- "n_children"
colnames(table8)[1] <- "CZ"

# Convert CZ to numeric for joining
table5$CZ <- as.numeric(table5$CZ)
table8$CZ <- as.numeric(table8$CZ)

```

Merge Tables and Apply Sample Restrictions

Following Chetty et al., we restrict to CZs with at least 250 children to ensure reliable mobility estimates.

```

# Merge mobility estimates with CZ characteristics
data <- table5 %>%
  select(CZ, RM, AM, n_children) %>%
  inner_join(table8, by = "CZ")

# Convert covariate columns to numeric
numeric_cols <- c("RM", "AM", "n_children", names(table8)[4:ncol(table8)])
for (col in numeric_cols) {
  if (col %in% names(data)) {
    data[[col]] <- as.numeric(data[[col]])
  }
}

# Apply sample restriction: n >= 250 children
data <- data %>%
  mutate(n250 = ifelse(n_children >= 250, 1, 0))

df <- data %>%
  filter(n250 == 1) %>%
  filter(!is.na(RM))

cat(sprintf("Final sample: %d CZs with n >= 250 children\n", nrow(df)))

```

Final sample: 709 CZs with n >= 250 children

```
cat(sprintf("(Dropped %d CZs with insufficient sample size)\n", nrow(data) - nrow(df)))
```

(Dropped 32 CZs with insufficient sample size)

Part 2: Permutation Test Implementation

The DiCiccio & Romano (2017) Permutation Regression Test

This function implements Theorem 3.2 from DiCiccio and Romano (2017). The key innovation is using a **studentized test statistic** that is asymptotically pivotal, meaning the permutation distribution approximates the true null distribution even without exchangeability.

The null hypothesis is $H_0 : \beta = 0$ (no linear relationship between X and Y).

```

#' Permutation Regression Test (DiCiccio & Romano 2017, Theorem 3.2)
#'
#' Tests H0: beta = 0 using a heteroskedasticity-robust Wald statistic.
#' The test permutes Y values to generate the null distribution.
#'

```

```

#' @param X Matrix of predictor variables (must be centered to mean zero)
#' @param Y Vector of response variable (must be centered to mean zero)
#' @param B Number of permutations (default: 10,000)
#' @param seed Random seed for reproducibility
#' @return List with observed statistic, permutation distribution, and p-value

perm_test_regression <- function(X, Y, B = 10000, seed = NULL) {
  if (!is.null(seed)) set.seed(seed)

  X <- as.matrix(X)
  Y <- as.numeric(Y)
  n <- nrow(X)

  if (n != length(Y)) stop("X and Y must have same number of rows")

  # Always include an intercept (per Remark 3.1 of the paper)
  X <- cbind(1, X)

  # Compute the heteroskedasticity-robust Wald statistic S_n
  compute_Sn <- function(X, Y) {
    XtX <- crossprod(X)           # X'X
    XtY <- crossprod(X, Y)        # X'Y
    beta_hat <- solve(XtX, XtY)    # OLS estimate
    resid <- as.vector(Y - X %*% beta_hat)

    Sigma_hat <- XtX / n           # (1/n) X'X
    r2 <- resid^2
    Omega_hat <- crossprod(X, X * r2) / n # HCO robust covariance

    tmp <- solve(Sigma_hat, beta_hat)
    as.numeric(n * t(tmp) %*% Omega_hat %*% tmp)
  }

  # Observed test statistic
  S_obs <- compute_Sn(X, Y)

  # Generate permutation distribution under H0
  S_perm <- replicate(B, {
    Y_perm <- sample(Y, replace = FALSE) # Permute Y values
    compute_Sn(X, Y_perm)
  })

  # Compute p-value (proportion of permuted stats >= observed)
  # +1 correction ensures valid p-value (Phipson & Smyth, 2010)
  pval <- (sum(S_perm >= S_obs) + 1) / (B + 1)

  list(S_obs = S_obs, S_perm = S_perm, p_value = pval, B = B)
}

```

Part 3: Define Covariates from Table VIII

The 35 Covariates

These are the same covariates analyzed in Online Appendix Table VIII of Chetty et al. (2014), organized by thematic category.

```

# Column names match R's processing of CSV headers
# (spaces -> dots, special characters -> dots)
covariates_to_test <- list(

  # RACE (1 variable)
  Race = c(
    "Frac..Black"          # Fraction Black Residents
  ),

  # SEGREGATION (5 variables)
  Segregation = c(
    "Racial.Segregation",  # Racial Segregation (Theil Index)
    "Income.Segregation",  # Income Segregation (Theil Index)
    "Segregation.of.Poverty...p25.", # Segregation of Poverty (<p25)
    "Segregation.of.Affluence...p75.", # Segregation of Affluence (>p75)
    "Frac..with.Commute...15.Mins"    # Share with Commute < 15 Mins
  ),

  # INCOME DISTRIBUTION (5 variables)
  Income_Distribution = c(
    "Household.Income.per.capita", # Household Income per Capita
    "Gini",                        # Gini Coefficient
    "Top.1..Income.Share",         # Top 1% Income Share
    "Gini.Bottom.99.",             # Gini Bottom 99%
    "Frac..Between.p25.and.p75"    # Fraction Middle Class
  ),

  # K-12 EDUCATION (4 variables)
  K12_Education = c(
    "School.Expenditure.per.Student", # School Expenditure per Student
    "Teacher.Student.Ratio",          # Teacher-Student Ratio
    "Test.Score.Percentile..Income.Adjusted", # Test Scores (Income Adjusted)
    "High.School.Dropout.Rate..Income.Adjusted." # Dropout Rate (Income Adjusted)
  ),

  # SOCIAL CAPITAL (3 variables)
  Social_Capital = c(
    "Social.Capital.Index",          # Social Capital Index
    "Fraction.Religious",            # Fraction Religious
    "Violent.Crime.Rate"             # Violent Crime Rate
  ),

  # FAMILY STRUCTURE (3 variables)
  Family_Structure = c(
    "Fraction.of.Children.with.Single.Mothers", # Fraction Single Mothers
    "Fraction.of.Adults.Divorced",             # Fraction Divorced
    "Fraction.of.Adults.Married"               # Fraction Married
  ),

  # TAX (4 variables)
  Tax = c(
    "Local.Tax.Rate",                # Local Tax Rate
    "Local.Government.Expenditures.Per.Capita", # Local Gov Expenditures
    "State.EITC.Exposure",           # State EITC Exposure
    "State.Income.Tax.Progressivity"  # State Income Tax Progressivity
  ),

```

```

# COLLEGE (3 variables)
College = c(
  "Number.of.Colleges.per.Capita",      # Colleges per Capita
  "College.Tuition",                   # Mean College Tuition
  "College.Graduation.Rate..Income.Adjusted." # Graduation Rate (Inc. Adj.)
),

# LOCAL LABOR MARKET (4 variables)
Local_Labor_Market = c(
  "Labor.Force.Participation.Rate",      # Labor Force Participation
  "Manufacturing.Employment.Share",      # Manufacturing Employment Share
  "Growth.in.Chinese.Imports.1990.2000", # Growth in Chinese Imports
  "Teenage.Labor.Force.Participation.Rate" # Teen Labor Force Participation
),

# MIGRATION (3 variables)
Migration = c(
  "Migration.Inflow.Rate",              # Migration Inflow Rate
  "Migration.Outflow.Rate",            # Migration Outflow Rate
  "Frac..Foreign.Born"                  # Fraction Foreign Born
)

all_covariates <- unlist(covariates_to_test, use.names = FALSE)
cat(sprintf("Testing %d covariates across %d categories\n",
  length(all_covariates), length(covariates_to_test)))

```

Testing 35 covariates across 10 categories

Part 4: Classical OLS Replication

Replicating Table VIII Results

This phase replicates Chetty et al.'s classical OLS results. For each covariate, we:

1. Standardize both the covariate and relative mobility (mean 0, SD 1)
2. Regress relative mobility on the covariate
3. The standardized coefficient equals the Pearson correlation

```

# Named vector mapping R-mangled column names to clean display names.
# Defined once and reused for both diagnostics and results tables.
covariate_name_map <- c(
  "Frac..Black" = "Fraction Black Residents",
  "Racial.Segregation" = "Racial Segregation",
  "Income.Segregation" = "Income Segregation",
  "Segregation.of.Poverty...p25." = "Segregation of Poverty (<p25)",
  "Segregation.of.Affluence...p75." = "Segregation of Affluence (>p75)",
  "Frac..with.Commute...15.Mins" = "Share with Commute < 15 Mins",
  "Household.Income.per.capita" = "Household Income per Capita",
  "Gini" = "Gini Coefficient",
  "Top.1..Income.Share" = "Top 1% Income Share",
  "Gini.Bottom.99." = "Gini Bottom 99%",
  "Frac..Between.p25.and.p75" = "Fraction Middle Class",
  "School.Expenditure.per.Student" = "School Expenditure per Student",
  "Teacher.Student.Ratio" = "Teacher-Student Ratio",
  "Test.Score.Percentile..Income.Adjusted" = "Test Score Percentile (Inc. Adj.)",
  "High.School.Dropout.Rate..Income.Adjusted." = "HS Dropout Rate (Inc. Adj.)",
  "Social.Capital.Index" = "Social Capital Index",

```

```

"Fraction.Religious" = "Fraction Religious",
"Violent.Crime.Rate" = "Violent Crime Rate",
"Fraction.of.Children.with.Single.Mothers" = "Fraction Single Mothers",
"Fraction.of.Adults.Divorced" = "Fraction Divorced",
"Fraction.of.Adults.Married" = "Fraction Married",
"Local.Tax.Rate" = "Local Tax Rate",
"Local.Government.Expenditures.Per.Capita" = "Local Gov Expenditures",
"State.EITC.Exposure" = "State EITC Exposure",
"State.Income.Tax.Progressivity" = "State Income Tax Progressivity",
"Number.of.Colleges.per.Capita" = "Colleges per Capita",
"College.Tuition" = "Mean College Tuition",
"College.Graduation.Rate..Income.Adjusted." = "College Grad Rate (Inc. Adj.)",
"Labor.Force.Participation.Rate" = "Labor Force Participation",
"Manufacturing.Employment.Share" = "Manufacturing Employment Share",
"Growth.in.Chinese.Imports.1990.2000" = "Growth in Chinese Imports",
"Teenage.Labor.Force.Participation.Rate" = "Teen Labor Force Participation",
"Migration.Inflow.Rate" = "Migration Inflow Rate",
"Migration.Outflow.Rate" = "Migration Outflow Rate",
"Frac..Foreign.Born" = "Fraction Foreign Born"
)

```

```

clean_covariate_name <- function(x) {
  ifelse(x %in% names(covariate_name_map),
    covariate_name_map[x],
    gsub("\\.", " ", x))
}

```

```

diagnostics_fun <- function(model, X, Y) {
  r_pearson <- cor(X, Y, method = "pearson")
  r_spearman <- cor(X, Y, method = "spearman")
  data.frame(
    Pearson_r = r_pearson,
    Spearman_r = r_spearman,
    Abs_r_minus_rho = abs(r_pearson - r_spearman),
    BP_p_value = bptest(model)$p.value,
    White_p_value = bptest(model, ~ X + I(X^2))$p.value,
    JB_p_value = jarque.bera.test(residuals(model))$p.value,
    Max_Cooks_D = max(cooks.distance(model)),
    Max_Leverage = max(hatvalues(model))
  )
}

```

```

# Initialize results dataframe
results <- data.frame(
  category = character(),
  covariate = character(),
  n_obs = integer(),
  beta = numeric(), # Standardized coefficient (= Pearson r)
  se = numeric(), # Standard error
  t_stat = numeric(), # t-statistic
  p_classical = numeric(), # Classical p-value from summary(lm)
  p_cor_test = numeric(), # p-value from cor.test
  p_waldtest_hc0 = numeric(), # p-value from waldtest with HCO
  p_waldtest_hc3 = numeric(), # p-value from waldtest with HC3
  stringsAsFactors = FALSE
)

```

```

# Initialize diagnostics dataframe
diagnostics_results <- data.frame(
  category = character(),
  covariate = character(),
  Pearson_r = numeric(),
  Spearman_r = numeric(),
  Abs_r_minus_rho = numeric(),
  BP_p_value = numeric(),
  White_p_value = numeric(),
  JB_p_value = numeric(),
  Max_Cooks_D = numeric(),
  Max_Leverage = numeric(),
  stringsAsFactors = FALSE
)

# Loop through all covariates
for (category_name in names(covariates_to_test)) {
  cat(sprintf("\n%s:\n", toupper(category_name)))
  cat(paste(rep("-", 70), collapse = ""), "\n")

  for (covariate_name in covariates_to_test[[category_name]]) {

    # Check if covariate exists in data
    if (!(covariate_name %in% names(df))) {
      cat(sprintf("  %s: NOT FOUND\n", covariate_name))
      next
    }

    # Extract complete cases
    temp_data <- df %>%
      select(RM, all_of(covariate_name)) %>%
      drop_na()

    if (nrow(temp_data) < 50) {
      cat(sprintf("  %s: Insufficient data (n=%d)\n", covariate_name, nrow(temp_data)))
      next
    }

    n_obs <- nrow(temp_data)

    # Standardize variables (mean 0, SD 1)
    y <- temp_data$RM
    y_std <- (y - mean(y)) / sd(y)
    x <- temp_data[[covariate_name]]
    x_std <- (x - mean(x)) / sd(x)

    # Fit OLS regression
    fit <- lm(y_std ~ x_std)

    # Compute regression diagnostics (reuses fit)
    diag_row <- diagnostics_fun(fit, x_std, y_std)
    diag_row$category <- category_name
    diag_row$covariate <- covariate_name
    diagnostics_results <- rbind(diagnostics_results, diag_row)
    fit_summary <- summary(fit)
    coef_table <- coef(fit_summary)
  }
}

```

```

beta <- coef_table[2, "Estimate"]
se <- coef_table[2, "Std. Error"]
t_stat <- coef_table[2, "t value"]
p_classical <- coef_table[2, "Pr(>|t|)"]

# --- cor.test (non-permutation correlation test) ---
cor_result <- cor.test(x = x_std, y = y_std,
                      alternative = "two.sided",
                      method = "pearson",
                      conf.level = 0.95)
p_cor_test <- cor_result$p.value

# --- waldtest with HCO (heteroskedasticity-robust) ---
waldtest_hc0 <- waldtest(fit, vcov = vcovHC(fit, type = "HCO"))
p_waldtest_hc0 <- waldtest_hc0$`Pr(>F)`[2]

# --- waldtest with HC3 (heteroskedasticity-robust, small-sample) ---
waldtest_hc3 <- waldtest(fit, vcov = vcovHC(fit, type = "HC3"))
p_waldtest_hc3 <- waldtest_hc3$`Pr(>F)`[2]

# Store results
results <- rbind(results, data.frame(
  category = category_name,
  covariate = covariate_name,
  n_obs = n_obs,
  beta = beta,
  se = se,
  t_stat = t_stat,
  p_classical = p_classical,
  p_cor_test = p_cor_test,
  p_waldtest_hc0 = p_waldtest_hc0,
  p_waldtest_hc3 = p_waldtest_hc3
))

# Print result
sig_icon <- ifelse(p_classical < 0.05, "*", " ")
cat(sprintf(" %-35s n=%3d b=%6.3f p=%5.3f%s\n",
  substr(covariate_name, 1, 35), n_obs, beta, p_classical, sig_icon))
}
}

```

RACE:

```

-----
Frac..Black                n=709  b= 0.631  p=0.000*

```

SEGREGATION:

```

-----
Racial.Segregation        n=709  b= 0.406  p=0.000*
Income.Segregation        n=709  b= 0.183  p=0.000*
Segregation.of.Poverty...p25. n=709  b= 0.218  p=0.000*
Segregation.of.Affluence...p75. n=709  b= 0.146  p=0.000*
Frac..with.Commute...15.Mins n=709  b=-0.447  p=0.000*

```

INCOME_DISTRIBUTION:

```

-----
Household.Income.per.capita n=709  b=-0.145  p=0.000*
Gini                       n=709  b= 0.346  p=0.000*

```

Top.1..Income.Share	n=709	b= 0.019	p=0.617
Gini.Bottom.99.	n=709	b= 0.473	p=0.000*
Frac..Between.p25.and.p75	n=709	b=-0.451	p=0.000*

K12_EDUCATION:

School.Expenditure.per.Student	n=701	b=-0.279	p=0.000*
Teacher.Student.Ratio	n=680	b= 0.009	p=0.805
Test.Score.Percentile..Income.Adjus	n=705	b=-0.317	p=0.000*
High.School.Dropout.Rate..Income.Ad	n=593	b= 0.328	p=0.000*

SOCIAL_CAPITAL:

Social.Capital.Index	n=693	b=-0.327	p=0.000*
Fraction.Religious	n=709	b=-0.101	p=0.007*
Violent.Crime.Rate	n=682	b= 0.217	p=0.000*

FAMILY_STRUCTURE:

Fraction.of.Children.with.Single.Mo	n=709	b= 0.641	p=0.000*
Fraction.of.Adults.Divorced	n=709	b= 0.158	p=0.000*
Fraction.of.Adults.Married	n=709	b=-0.370	p=0.000*

TAX:

Local.Tax.Rate	n=708	b=-0.328	p=0.000*
Local.Government.Expenditures.Per.C	n=707	b=-0.301	p=0.000*
State.EITC.Exposure	n=709	b=-0.144	p=0.000*
State.Income.Tax.Progressivity	n=709	b=-0.150	p=0.000*

COLLEGE:

Number.of.Colleges.per.Capita	n=584	b=-0.125	p=0.002*
College.Tuition	n=580	b= 0.109	p=0.008*
College.Graduation.Rate..Income.Adj	n=581	b=-0.025	p=0.553

LOCAL_LABOR_MARKET:

Labor.Force.Participation.Rate	n=709	b=-0.237	p=0.000*
Manufacturing.Employment.Share	n=709	b= 0.393	p=0.000*
Growth.in.Chinese.Imports.1990.2000	n=693	b= 0.171	p=0.000*
Teenage.Labor.Force.Participation.R	n=709	b=-0.516	p=0.000*

MIGRATION:

Migration.Inflow.Rate	n=707	b=-0.084	p=0.025*
Migration.Outflow.Rate	n=707	b=-0.150	p=0.000*
Frac..Foreign.Born	n=709	b=-0.247	p=0.000*

Summary

```
cat(sprintf("\n\nCLASSICAL OLS SUMMARY\n"))
```

CLASSICAL OLS SUMMARY

```
cat(paste(rep("=", 70), collapse = ""), "\n")
```

```
=====
```

```
cat(sprintf("Total covariates tested: %d\n", nrow(results)))
```

Total covariates tested: 35

```
cat(sprintf("Significant at alpha=0.05: %d (%.1f%%)\n",
  sum(results$p_classical < 0.05),
  100 * mean(results$p_classical < 0.05)))
```

Significant at alpha=0.05: 32 (91.4%)

Regression Diagnostics

This table reports diagnostic statistics for each covariate regression on standardized data. Large $|r - \rho|$ suggests non-linearity. Small Breusch-Pagan or White p-values indicate heteroskedasticity. Small Jarque-Bera p-values indicate non-normal residuals. High Cook's D or leverage flags influential observations.

```
# Add clean covariate names
diagnostics_results <- diagnostics_results %>%
  mutate(covariate_clean = clean_covariate_name(covariate))

# Display table
knitr::kable(
  diagnostics_results %>%
    select(category, covariate_clean, Pearson_r, Spearman_r, Abs_r_minus_rho,
           BP_p_value, White_p_value, JB_p_value, Max_Cooks_D, Max_Leverage),
  col.names = c("Category", "Covariate", "Pearson r", "Spearman r",
                "|r - rho|", "BP p", "White p", "JB p",
                "Max Cook's D", "Max Leverage"),
  digits = 4,
  caption = "Regression Diagnostics for 35 Covariate Regressions on Relative Mobility"
)
```

Table 3: Regression Diagnostics for 35 Covariate Regressions on Relative Mobility

	Category	Covariate	Pearson r	Spearman r	$ r - \rho $	BP p	White p	JB p	Max Cook's D	Max Lever- age
BP	Race	Fraction Black Residents	0.6308	0.6841	0.0533	0.0000	0.0000	0.0000	0.0539	0.0320
BP1	Segregation	Racial Segregation	0.4060	0.4423	0.0363	0.0301	0.0357	0.0634	0.0425	0.0268
BP2	Segregation	Income Segregation	0.1831	0.2126	0.0295	0.0017	0.0020	0.0665	0.0262	0.0147
BP3	Segregation	Segregation of Poverty (<p25)	0.2177	0.2367	0.0190	0.0006	0.0007	0.0905	0.0263	0.0153
BP4	Segregation	Segregation of Affluence (>p75)	0.1457	0.1752	0.0296	0.0027	0.0030	0.0485	0.0249	0.0155
BP5	Segregation	Share with Commute < 15 Mins	-	-	0.0224	0.0001	0.0000	0.0061	0.1246	0.0189
BP6	Income_Distribution	Bottom 10% Household Income per Capita	-	-	0.0201	0.0030	0.0024	0.0003	0.0512	0.0301
BP7	Income_Distribution	Gini Coefficient	0.1453	0.1654	0.0525	0.0000	0.0000	0.0000	0.2090	0.0446
BP8	Income_Distribution	Top 10% Income Share	0.3455	0.3980	0.1571	0.0250	0.0003	0.0056	0.9797	0.1627
BP9	Income_Distribution	Bottom 99% Fraction	0.0188	0.1759	0.0043	0.0978	0.0347	0.0000	0.0409	0.0272
BP10	Income_Distribution	Top 10% Middle Class	0.4727	0.4684	0.0062	0.0035	0.0000	0.0000	0.0873	0.0175
			0.4511	0.4573						

Category	Covariate	Pearson r	Spearman r	r - rho	BP p	White p	JB p	Max Cook's D	Max Leverage
BP11	K12_Education_School Expenditure per Student	-0.2785	-0.3125	0.0340	0.0034	0.0052	0.0000	0.0183	0.0397
BP12	K12_Education_Teacher-Student Ratio	0.0095	0.0449	0.0355	0.7299	0.3376	0.0252	0.0271	0.0349
BP13	K12_Education_Test Score Percentile (Inc. Adj.)	-0.3170	-0.3329	0.0160	0.0004	0.0000	0.0000	0.1994	0.0235
BP14	K12_Education_HS Dropout Rate (Inc. Adj.)	0.3276	0.3486	0.0210	0.3047	0.0034	0.0050	0.1732	0.0426
BP15	Social_Capital_Social Capital Index	-0.3274	-0.3306	0.0032	0.6699	0.0159	0.0006	0.0897	0.0267
BP16	Social_Capital_Fraction Religious	-0.1008	-0.0527	0.0481	0.2780	0.0002	0.0185	0.1008	0.0158
BP17	Social_Capital_Violent Crime Rate	0.2174	0.3303	0.1129	0.0000	0.0000	0.0008	12.2806	0.4868
BP18	Family_Structure_Fraction Single Mothers	0.6406	0.6074	0.0332	0.0370	0.0620	0.0001	0.0598	0.0292
BP19	Family_Structure_Fraction Divorced	0.1581	0.1114	0.0468	0.0000	0.0000	0.2762	0.1090	0.0182
BP20	Family_Structure_Fraction Married	-0.3697	-0.3621	0.0076	0.0101	0.0015	0.0000	0.0909	0.0280
BP21	Tax_Local Tax Rate	-0.3280	-0.3333	0.0053	0.5929	0.3609	0.0022	0.1306	0.0578
BP22	Tax_Local Gov Expenditures	0.3013	0.3277	0.0264	0.0280	0.0000	0.0002	1.5014	0.2034
BP23	Tax_State EITC Exposure	0.1437	0.2172	0.0734	0.0017	0.0037	0.0002	0.0331	0.0393
BP24	Tax_State Income Tax Progressivity	0.1503	0.1535	0.0032	0.3140	0.0326	0.0088	0.0425	0.0283
BP25	College_Colleges per Capita	0.1251	0.0439	0.0812	0.0003	0.0002	0.0633	0.3053	0.1784
BP26	College_Mean College Tuition	0.1092	0.1394	0.0302	0.0008	0.0002	0.0525	0.0293	0.0527
BP27	College_College Grad Rate (Inc. Adj.)	0.0246	0.0255	0.0009	0.2234	0.4678	0.0061	0.0437	0.0277
BP28	Local_Labor_Market Participation	0.2365	0.2403	0.0037	0.0306	0.0033	0.0004	0.0482	0.0272
BP29	Local_Labor_Manufacturing Employment Share	0.3928	0.4153	0.0225	0.0947	0.0004	0.0086	0.1060	0.0195
BP30	Local_Labor_Chinese in Chinese Imports	0.1705	0.3312	0.1606	0.0523	0.0000	0.0017	1.9780	0.2608
BP31	Local_Labor_Market Participation	0.5160	0.5162	0.0002	0.8389	0.0339	0.0000	0.0237	0.0094
BP32	Migration_Migration Inflow Rate	0.0844	0.0670	0.0174	0.0000	0.0000	0.0002	0.0442	0.0493
BP33	Migration_Migration Outflow Rate	0.1500	0.1521	0.0021	0.0116	0.0331	0.0003	0.0299	0.0329
BP34	Migration_Fraction Foreign Born	0.2473	0.2080	0.0393	0.0002	0.0000	0.0000	0.0375	0.0717

```
# Save to CSV
write.csv(diagnostics_results,
  "../results/main/regression_diagnostics.csv",
  row.names = FALSE)
cat("Saved: ../results/main/regression_diagnostics.csv\n")
```

Saved: ../results/main/regression_diagnostics.csv

Part 5: Permutation Test Comparison

Comparing Classical and Permutation Inference

Now we run both permutation tests (perk correlation and DiCiccio & Romano regression) for each covariate and compare to the classical results. This tests whether the significance conclusions are robust to the choice of inference method.

```
# Add permutation results columns
results$p_perk <- NA           # perk permutation correlation p-value
results$p_perm_reg <- NA       # DiCiccio & Romano permutation regression p-value
results$classical_sig <- results$p_classical < 0.05
results$perk_sig <- NA
results$perm_reg_sig <- NA
results$all_agree <- NA        # TRUE if classical, perk, perm_reg agree

# Loop through covariates
for (i in seq_len(nrow(results))) {
  covariate_name <- results$covariate[i]
  category_name <- results$category[i]

  if (i == 1 || results$category[i] != results$category[i - 1]) {
    cat(sprintf("\n%s:\n", toupper(category_name)))
    cat(paste(rep("-", 70), collapse = ""), "\n")
  }

  # Extract and standardize data
  temp_data <- df %>%
    select(RM, all_of(covariate_name)) %>%
    drop_na()

  y <- temp_data$RM
  y_std <- (y - mean(y)) / sd(y)
  x <- temp_data[[covariate_name]]
  x_std <- (x - mean(x)) / sd(x)

  # Center to mean zero (required for permutation test)
  x_m0 <- x_std - mean(x_std)
  y_m0 <- y_std - mean(y_std)

  # --- Method 2: perk permutation correlation test ---
  set.seed(42 + i)
  perk_result <- perk::perk_test(x_std, y_std, method = "pearson", alternative = "two.sided")
  p_perk <- perk_result$p.value
  results$p_perk[i] <- p_perk
  results$perk_sig[i] <- p_perk < 0.05

  # --- Method 3: DiCiccio & Romano permutation regression test ---
  perm_reg_result <- perm_test_regression(
    X = x_m0,
    Y = y_m0,
    B = 10000,
    seed = 42 + i
  )
}
```

```

p_perm_reg <- perm_reg_result$p_value
results$p_perm_reg[i] <- p_perm_reg
results$perm_reg_sig[i] <- p_perm_reg < 0.05

# Check if all three methods agree
results$all_agree[i] <- (results$classical_sig[i] == results$perk_sig[i]) &
                        (results$classical_sig[i] == results$perm_reg_sig[i])

# Print result
agree_icon <- ifelse(results$all_agree[i], "[OK]", "[X]")
sig_icon <- ifelse(results$classical_sig[i], "*", " ")

cat(sprintf("  %s %-25s b=%6.3f p_class=%5.3f p_perk=%5.3f p_reg=%5.3f%s\n",
  agree_icon,
  substr(covariate_name, 1, 25),
  results$beta[i],
  results$p_classical[i],
  p_perk,
  p_perm_reg,
  sig_icon))
}

```

RACE:

```

-----
[OK] Frac..Black          b= 0.631 p_class=0.000 p_perk=0.000 p_reg=0.000*

```

SEGREGATION:

```

-----
[OK] Racial.Segregation   b= 0.406 p_class=0.000 p_perk=0.000 p_reg=0.000*
[OK] Income.Segregation   b= 0.183 p_class=0.000 p_perk=0.000 p_reg=0.000*
[OK] Segregation.of.Poverty... b= 0.218 p_class=0.000 p_perk=0.000 p_reg=0.000*
[OK] Segregation.of.Affluence. b= 0.146 p_class=0.000 p_perk=0.000 p_reg=0.000*
[OK] Frac..with.Commute...15.M b=-0.447 p_class=0.000 p_perk=0.000 p_reg=0.000*

```

INCOME_DISTRIBUTION:

```

-----
[OK] Household.Income.per.capi b=-0.145 p_class=0.000 p_perk=0.000 p_reg=0.000*
[OK] Gini                    b= 0.346 p_class=0.000 p_perk=0.000 p_reg=0.000*
[OK] Top.1..Income.Share      b= 0.019 p_class=0.617 p_perk=0.741 p_reg=0.412
[OK] Gini.Bottom.99.          b= 0.473 p_class=0.000 p_perk=0.000 p_reg=0.000*
[OK] Frac..Between.p25.and.p75 b=-0.451 p_class=0.000 p_perk=0.000 p_reg=0.000*

```

K12_EDUCATION:

```

-----
[OK] School.Expenditure.per.St b=-0.279 p_class=0.000 p_perk=0.000 p_reg=0.000*
[OK] Teacher.Student.Ratio      b= 0.009 p_class=0.805 p_perk=0.814 p_reg=0.791
[OK] Test.Score.Percentile..In b=-0.317 p_class=0.000 p_perk=0.000 p_reg=0.000*
[OK] High.School.Dropout.Rate. b= 0.328 p_class=0.000 p_perk=0.000 p_reg=0.000*

```

SOCIAL_CAPITAL:

```

-----
[OK] Social.Capital.Index      b=-0.327 p_class=0.000 p_perk=0.000 p_reg=0.000*
[OK] Fraction.Religious        b=-0.101 p_class=0.007 p_perk=0.018 p_reg=0.002*
[OK] Violent.Crime.Rate        b= 0.217 p_class=0.000 p_perk=0.000 p_reg=0.000*

```

FAMILY_STRUCTURE:

[OK] Fraction.of.Children.with	b= 0.641	p_class=0.000	p_perk=0.000	p_reg=0.000*
[OK] Fraction.of.Adults.Divorc	b= 0.158	p_class=0.000	p_perk=0.001	p_reg=0.000*
[OK] Fraction.of.Adults.Marrie	b=-0.370	p_class=0.000	p_perk=0.000	p_reg=0.000*

TAX:

[OK] Local.Tax.Rate	b=-0.328	p_class=0.000	p_perk=0.000	p_reg=0.000*
[OK] Local.Government.Expendit	b=-0.301	p_class=0.000	p_perk=0.000	p_reg=0.000*
[OK] State.EITC.Exposure	b=-0.144	p_class=0.000	p_perk=0.000	p_reg=0.010*
[OK] State.Income.Tax.Progress	b=-0.150	p_class=0.000	p_perk=0.000	p_reg=0.000*

COLLEGE:

[OK] Number.of.Colleges.per.Ca	b=-0.125	p_class=0.002	p_perk=0.003	p_reg=0.001*
[OK] College.Tuition	b= 0.109	p_class=0.008	p_perk=0.004	p_reg=0.013*
[OK] College.Graduation.Rate..	b=-0.025	p_class=0.553	p_perk=0.520	p_reg=0.563

LOCAL_LABOR_MARKET:

[OK] Labor.Force.Participation	b=-0.237	p_class=0.000	p_perk=0.000	p_reg=0.000*
[OK] Manufacturing.Employment.	b= 0.393	p_class=0.000	p_perk=0.000	p_reg=0.000*
[OK] Growth.in.Chinese.Imports	b= 0.171	p_class=0.000	p_perk=0.000	p_reg=0.000*
[OK] Teenage.Labor.Force.Parti	b=-0.516	p_class=0.000	p_perk=0.000	p_reg=0.000*

MIGRATION:

[OK] Migration.Inflow.Rate	b=-0.084	p_class=0.025	p_perk=0.018	p_reg=0.042*
[OK] Migration.Outflow.Rate	b=-0.150	p_class=0.000	p_perk=0.000	p_reg=0.000*
[OK] Frac..Foreign.Born	b=-0.247	p_class=0.000	p_perk=0.000	p_reg=0.000*

Results Summary

Agreement Between Methods

```
cat("\n")
```

```
cat(paste(rep("=", 70), collapse = ""), "\n")
```

```
cat("RESULTS SUMMARY\n")
```

RESULTS SUMMARY

```
cat(paste(rep("=", 70), collapse = ""), "\n\n")
```

```
cat("=== NON-PERMUTATION METHODS ===\n\n")
```

=== NON-PERMUTATION METHODS ===

```
cat("1. CLASSICAL OLS - summary(lm)\n")
```

1. CLASSICAL OLS - summary(lm)

```
cat(sprintf(" - Covariates tested: %d\n", nrow(results)))
```

- Covariates tested: 35

```
cat(sprintf("    - Significant (p < 0.05): %d (0.1f%%)\n",
  sum(results$classical_sig),
  100 * mean(results$classical_sig)))
```

- Significant (p < 0.05): 32 (91.4%)

```
cat("\n2. COR.TEST (Pearson correlation)\n")
```

2. COR.TEST (Pearson correlation)

```
cor_test_sig <- results$p_cor_test < 0.05
cat(sprintf("    - Significant (p < 0.05): %d (0.1f%%)\n",
  sum(cor_test_sig),
  100 * mean(cor_test_sig)))
```

- Significant (p < 0.05): 32 (91.4%)

```
cat("\n3. WALDTEST with HCO (heteroskedasticity-robust)\n")
```

3. WALDTEST with HCO (heteroskedasticity-robust)

```
hc0_sig <- results$p_waldtest_hc0 < 0.05
cat(sprintf("    - Significant (p < 0.05): %d (0.1f%%)\n",
  sum(hc0_sig),
  100 * mean(hc0_sig)))
```

- Significant (p < 0.05): 32 (91.4%)

```
cat("\n4. WALDTEST with HC3 (heteroskedasticity-robust, small-sample)\n")
```

4. WALDTEST with HC3 (heteroskedasticity-robust, small-sample)

```
hc3_sig <- results$p_waldtest_hc3 < 0.05
cat(sprintf("    - Significant (p < 0.05): %d (0.1f%%)\n",
  sum(hc3_sig),
  100 * mean(hc3_sig)))
```

- Significant (p < 0.05): 31 (88.6%)

```
cat("\n=== PERMUTATION METHODS ===\n\n")
```

=== PERMUTATION METHODS ===

```
cat("5. PERK PERMUTATION CORRELATION\n")
```

5. PERK PERMUTATION CORRELATION

```
cat(sprintf("    - Significant (p < 0.05): %d (0.1f%%)\n",
  sum(results$perk_sig),
  100 * mean(results$perk_sig)))
```

- Significant (p < 0.05): 32 (91.4%)

```
cat("\n6. PERMUTATION REGRESSION (DiCiccio & Romano 2017)\n")
```

6. PERMUTATION REGRESSION (DiCiccio & Romano 2017)

```
cat(sprintf("    - Significant (p < 0.05): %d (0.1f%%)\n",
  sum(results$perm_reg_sig),
  100 * mean(results$perm_reg_sig)))
```

```
    - Significant (p < 0.05): 32 (91.4%)
```

```
cat("\n=== AGREEMENT ANALYSIS ===\n\n")
```

```
=== AGREEMENT ANALYSIS ===
```

```
cat(sprintf("All six methods agree: %d/%d (0.1f%%)\n",
  sum(results$all_agree & cor_test_sig == results$classical_sig &
    hc0_sig == results$classical_sig & hc3_sig == results$classical_sig),
  nrow(results),
  100 * mean(results$all_agree & cor_test_sig == results$classical_sig &
    hc0_sig == results$classical_sig & hc3_sig == results$classical_sig)))
```

```
All six methods agree: 34/35 (97.1%)
```

```
# Show any disagreements (check all 6 methods, not just the 3 in all_agree)
```

```
all_six_agree <- results$all_agree &
  cor_test_sig == results$classical_sig &
  hc0_sig == results$classical_sig &
  hc3_sig == results$classical_sig
```

```
disagreements <- results %>% filter(!all_six_agree)
```

```
if (nrow(disagreements) > 0) {
  cat(sprintf("\nDISAGREEMENTS (%d):\n", nrow(disagreements)))
  for (j in seq_len(nrow(disagreements))) {
    row <- disagreements[j, ]
    cat(sprintf("    - %s:\n", row$covariate))
    cat(sprintf("        summary(lm) p=%.4f (%s)\n",
      row$p_classical, ifelse(row$classical_sig, "SIG", "NOT SIG")))
    cat(sprintf("        cor.test p=%.4f (%s)\n",
      row$p_cor_test, ifelse(row$p_cor_test < 0.05, "SIG", "NOT SIG")))
    cat(sprintf("        waldtest HCO p=%.4f (%s)\n",
      row$p_waldtest_hc0, ifelse(row$p_waldtest_hc0 < 0.05, "SIG", "NOT SIG")))
    cat(sprintf("        waldtest HC3 p=%.4f (%s)\n",
      row$p_waldtest_hc3, ifelse(row$p_waldtest_hc3 < 0.05, "SIG", "NOT SIG")))
    cat(sprintf("        perk p=%.4f (%s)\n",
      row$p_perk, ifelse(row$perk_sig, "SIG", "NOT SIG")))
    cat(sprintf("        perm_reg p=%.4f (%s)\n",
      row$p_perm_reg, ifelse(row$perm_reg_sig, "SIG", "NOT SIG")))
  }
} else {
  cat("\nNo disagreements - all methods agree on significance!\n")
}
```

```
DISAGREEMENTS (1):
```

```
- Violent.Crime.Rate:
  summary(lm) p=0.0000 (SIG)
  cor.test p=0.0000 (SIG)
  waldtest HCO p=0.0298 (SIG)
  waldtest HC3 p=0.2473 (NOT SIG)
  perk p=0.0000 (SIG)
  perm_reg p=0.0001 (SIG)
```

```
cat("\n=== P-VALUE CORRELATIONS ===\n\n")
```

```
=== P-VALUE CORRELATIONS ===
```

```
cat("Correlation methods:\n")
```

```
Correlation methods:
```

```
cat(sprintf("    - cor.test vs perk: r = %.3f\n",  
  cor(results$p_cor_test, results$p_perk)))
```

```
    - cor.test vs perk: r = 0.995
```

```
cat("\nRegression methods:\n")
```

```
Regression methods:
```

```
cat(sprintf("    - summary(lm) vs waldtest HC0: r = %.3f\n",  
  cor(results$p_classical, results$p_waldtest_hc0)))
```

```
    - summary(lm) vs waldtest HC0: r = 0.995
```

```
cat(sprintf("    - summary(lm) vs waldtest HC3: r = %.3f\n",  
  cor(results$p_classical, results$p_waldtest_hc3)))
```

```
    - summary(lm) vs waldtest HC3: r = 0.973
```

```
cat(sprintf("    - summary(lm) vs perm_reg: r = %.3f\n",  
  cor(results$p_classical, results$p_perm_reg)))
```

```
    - summary(lm) vs perm_reg: r = 0.985
```

```
cat(sprintf("    - waldtest HC0 vs perm_reg: r = %.3f\n",  
  cor(results$p_waldtest_hc0, results$p_perm_reg)))
```

```
    - waldtest HC0 vs perm_reg: r = 0.965
```

```
cat(sprintf("    - waldtest HC3 vs perm_reg: r = %.3f\n",  
  cor(results$p_waldtest_hc3, results$p_perm_reg)))
```

```
    - waldtest HC3 vs perm_reg: r = 0.939
```

Non-Significant Covariates

These covariates were NOT significantly associated with relative mobility under any method:

```
nonsig <- results %>%  
  filter(!classical_sig & !perk_sig & !perm_reg_sig &  
    p_cor_test >= 0.05 & p_waldtest_hc0 >= 0.05 & p_waldtest_hc3 >= 0.05) %>%  
  select(category, covariate, beta, p_classical, p_cor_test,  
    p_waldtest_hc0, p_waldtest_hc3, p_perk, p_perm_reg)  
  
if (nrow(nonsig) > 0) {  
  cat("\nCovariates NOT significant under ALL six methods:\n\n")  
  for (j in seq_len(nrow(nonsig))) {  
    cat(sprintf("    - %s (b=%.3f)\n", nonsig$covariate[j], nonsig$beta[j]))  
    cat(sprintf("      Non-perm: lm=%.3f, cor.test=%.3f, HC0=%.3f, HC3=%.3f\n",  
      nonsig$p_classical[j], nonsig$p_cor_test[j],  
      nonsig$p_waldtest_hc0[j], nonsig$p_waldtest_hc3[j]))  
    cat(sprintf("      Perm:      perk=%.3f, perm_reg=%.3f\n",
```

```

        nonsig$p_perk[j], nonsig$p_perm_reg[j]))
    }
  } else {
    cat("\nAll covariates were significant under at least one method.\n")
  }
}

```

Covariates NOT significant under ALL six methods:

- Top.1..Income.Share (b=0.019)
Non-perm: lm=0.617, cor.test=0.617, HC0=0.746, HC3=0.775
Perm: perk=0.741, perm_reg=0.412
- Teacher.Student.Ratio (b=0.009)
Non-perm: lm=0.805, cor.test=0.805, HC0=0.817, HC3=0.819
Perm: perk=0.814, perm_reg=0.791
- College.Graduation.Rate..Income.Adjusted. (b=-0.025)
Non-perm: lm=0.553, cor.test=0.553, HC0=0.540, HC3=0.543
Perm: perk=0.520, perm_reg=0.563

Save Results

```

# Create output directory if needed
if (!dir.exists("../results/main")) {
  dir.create("../results/main", recursive = TRUE)
}

# Clean up covariate names for readability
results_clean <- results %>%
  mutate(covariate_clean = clean_covariate_name(covariate))

# Save results
write.csv(results_clean,
  "../results/main/mobility_test_results.csv",
  row.names = FALSE)
cat("Saved: ../results/main/mobility_test_results.csv\n")

```

Saved: ../results/main/mobility_test_results.csv

```

# Save workspace
save(results, results_clean, diagnostics_results, df, covariates_to_test,
  file = "../results/main/mobility_analysis_workspace.RData")
cat("Saved: ../results/main/mobility_analysis_workspace.RData\n")

```

Saved: ../results/main/mobility_analysis_workspace.RData

Visualization

Coefficient Plot

This plot shows the standardized regression coefficients for all 35 covariates, with colors indicating significance agreement across all six methods.

```

# Prepare data for plotting
plot_data <- results_clean %>%
  mutate(
    sig_status = case_when(
      classical_sig & perk_sig & perm_reg_sig &

```

```

      (p_cor_test < 0.05) & (p_waldtest_hc0 < 0.05) & (p_waldtest_hc3 < 0.05) ~ "All
↪ significant",
      !classical_sig & !perk_sig & !perm_reg_sig &
      (p_cor_test >= 0.05) & (p_waldtest_hc0 >= 0.05) & (p_waldtest_hc3 >= 0.05) ~ "None
↪ significant",
      TRUE ~ "Disagreement"
    )
  ) %>%
  arrange(beta)

plot_data$covariate_clean <- factor(
  plot_data$covariate_clean,
  levels = plot_data$covariate_clean
)

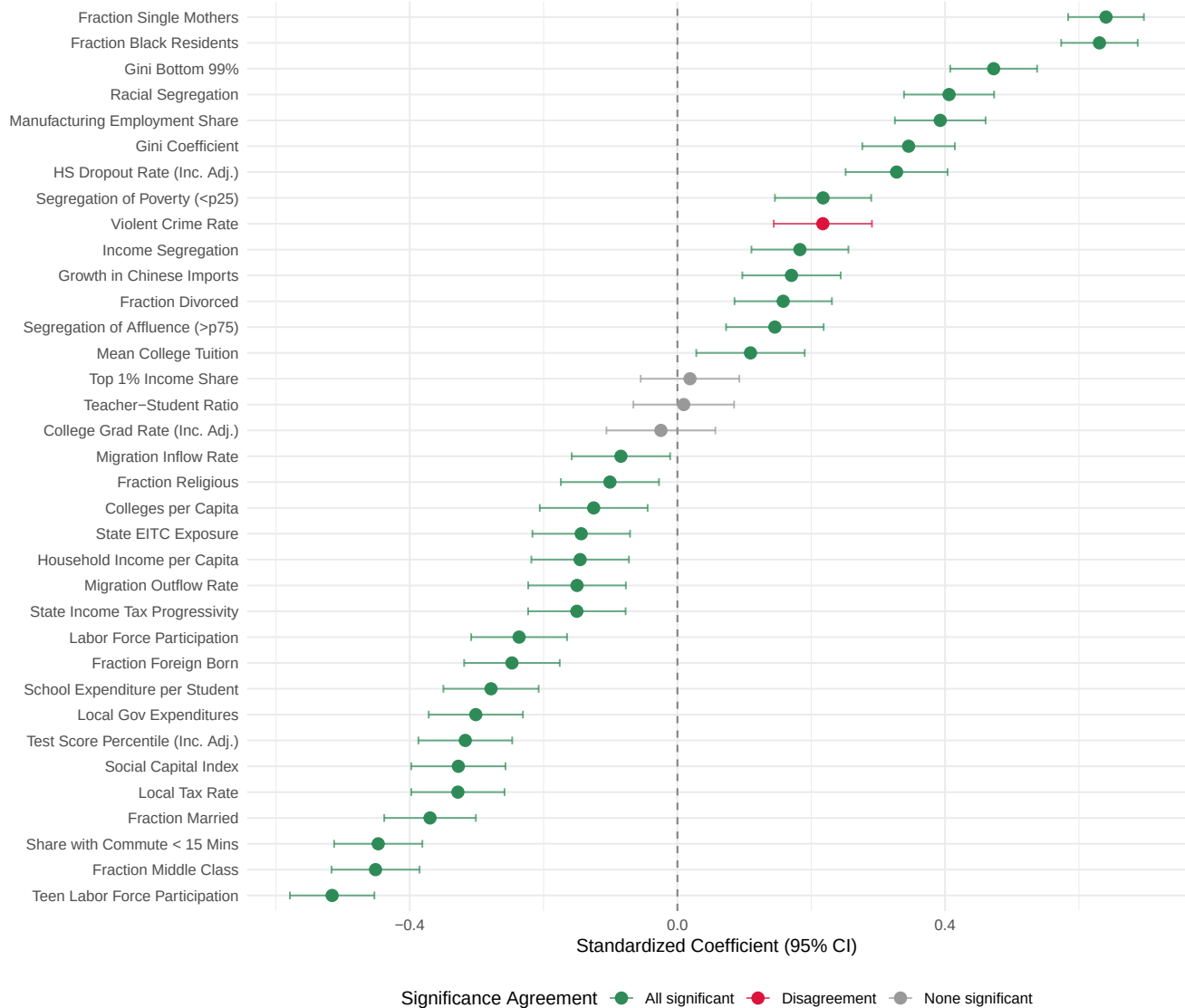
# Create coefficient plot
p <- ggplot(plot_data, aes(x = beta, y = covariate_clean)) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
  geom_errorbar(
    aes(xmin = beta - 1.96 * se, xmax = beta + 1.96 * se, color = sig_status),
    height = 0.3, alpha = 0.7
  ) +
  geom_point(aes(color = sig_status), size = 3) +
  scale_color_manual(values = c(
    "All significant" = "#2E8B57",
    "None significant" = "gray60",
    "Disagreement" = "#DC143C"
  )) +
  labs(
    title = "Predictors of Relative Mobility: Classical vs Permutation Inference",
    subtitle = "Replication of Chetty et al. (2014) Table VIII with robustness check (6
↪ methods)",
    x = "Standardized Coefficient (95% CI)",
    y = "",
    color = "Significance Agreement"
  ) +
  theme_minimal() +
  theme(
    legend.position = "bottom",
    axis.text.y = element_text(size = 9),
    plot.title = element_text(face = "bold"),
    plot.subtitle = element_text(color = "gray40")
  )

print(p)

```

Predictors of Relative Mobility: Classical vs Permutation Inference

Replication of Chetty et al. (2014) Table VIII with robustness check (6 methods)



```
ggsave("../results/main/coefficient_plot.png", p, width = 10, height = 9, dpi = 300)
cat("Saved: ../results/main/coefficient_plot.png\n")
```

Saved: ../results/main/coefficient_plot.png

Conclusion

```
cat(paste(rep("=", 70), collapse = ""), "\n")
```

```
cat("CONCLUSION\n")
```

CONCLUSION

```
cat(paste(rep("=", 70), collapse = ""), "\n\n")
```

```

# Check agreement across all 6 methods
cor_test_sig <- results$p_cor_test < 0.05
hc0_sig <- results$p_waldtest_hc0 < 0.05
hc3_sig <- results$p_waldtest_hc3 < 0.05

all_six_agree <- results$all_agree &
  cor_test_sig == results$classical_sig &
  hc0_sig == results$classical_sig &
  hc3_sig == results$classical_sig

agreement_pct <- 100 * mean(all_six_agree)

if (agreement_pct == 100) {
  cat("All 35 covariates show IDENTICAL significance conclusions under all\n")
  cat("six methods:\n")
  cat("  Non-permutation: summary(lm), cor.test, waldtest HC0, waldtest HC3\n")
  cat("  Permutation: perk, DiCiccio & Romano regression\n\n")
  cat("This demonstrates that Chetty et al.'s findings are ROBUST to the\n")
  cat("choice of inference method and do not depend on asymptotic normality\n")
  cat("assumptions.\n")
} else {
  cat(sprintf("%.1f%% of covariates show identical significance conclusions across\n",
    ↪ agreement_pct))
  cat("all six methods.\n\n")
  cat(sprintf("%d covariates show disagreement between at least two methods.\n",
    ↪ sum(!all_six_agree)))
}

```

97.1% of covariates show identical significance conclusions across all six methods.

1 covariates show disagreement between at least two methods.

```
cat(sprintf("\n\nKey Statistics:\n"))
```

Key Statistics:

```
cat(sprintf("  - Sample size: %d Commuting Zones\n", nrow(df)))
```

- Sample size: 709 Commuting Zones

```
cat(sprintf("  - Covariates tested: %d\n", nrow(results)))
```

- Covariates tested: 35

```
cat("\n  Non-permutation methods:\n")
```

Non-permutation methods:

```
cat(sprintf("    - summary(lm) significant: %d (%.1f%%)\n",
  sum(results$classical_sig), 100 * mean(results$classical_sig)))
```

- summary(lm) significant: 32 (91.4%)

```
cat(sprintf("    - cor.test significant: %d (%.1f%%)\n",
  sum(cor_test_sig), 100 * mean(cor_test_sig)))
```

```

- cor.test significant: 32 (91.4%)
cat(sprintf("    - waldtest HC0 significant: %d (0.1f%%)\n",
  sum(hc0_sig), 100 * mean(hc0_sig)))

- waldtest HC0 significant: 32 (91.4%)
cat(sprintf("    - waldtest HC3 significant: %d (0.1f%%)\n",
  sum(hc3_sig), 100 * mean(hc3_sig)))

- waldtest HC3 significant: 31 (88.6%)
cat("\n Permutation methods:\n")

Permutation methods:
cat(sprintf("    - perk significant: %d (0.1f%%)\n",
  sum(results$perk_sig), 100 * mean(results$perk_sig)))

- perk significant: 32 (91.4%)
cat(sprintf("    - perm_reg significant: %d (0.1f%%)\n",
  sum(results$perm_reg_sig), 100 * mean(results$perm_reg_sig)))

- perm_reg significant: 32 (91.4%)
cat("\n Key p-value correlations:\n")

Key p-value correlations:
cat(sprintf("    - cor.test vs perk: r = 0.3f\n",
  cor(results$p_cor_test, results$p_perk)))

- cor.test vs perk: r = 0.995
cat(sprintf("    - summary(lm) vs perm_reg: r = 0.3f\n",
  cor(results$p_classical, results$p_perm_reg)))

- summary(lm) vs perm_reg: r = 0.985
cat(sprintf("    - waldtest HC0 vs perm_reg: r = 0.3f\n",
  cor(results$p_waldtest_hc0, results$p_perm_reg)))

- waldtest HC0 vs perm_reg: r = 0.965

```

References

- Chetty, R., Hendren, N., Kline, P., & Saez, E. (2014). Where is the land of opportunity? The geography of intergenerational mobility in the United States. *Quarterly Journal of Economics*, 129(4), 1553-1623.
- DiCiccio, C. J., & Romano, J. P. (2017). Robust permutation tests for correlation and regression coefficients. *Journal of the American Statistical Association*, 112(519), 1211-1220.
- Jaimes, A. (2025). Where is the Land of Opportunity Replication. [Implementation of DiCiccio & Romano permutation tests]
- perk R package: Permutation tests for correlation coefficients. Available on CRAN.